

# ORF 245: Fundamentals of Statistics – Spring 2025

## Precept 6

### 1. Estimation

- (a) Parametric estimation: random sample  $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} f(x; \theta)$ .  $f$  known,  $\theta$  unknown. Target is to construct a point estimator  $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$  to estimate  $\theta$ .
- (b) Bias and unbiased estimator:  $\hat{\theta}$  is an unbiased estimator of  $\theta$  if  $\mathbb{E}_{X_1, \dots, X_n}[\hat{\theta}] = \theta$ . If  $\mathbb{E}[\hat{\theta}] \neq \theta$ ,  $\mathbb{E}[\hat{\theta}] - \theta$  is the bias of the estimator.
- (c) Evaluation of point estimator
  - Bias-and-variance decomposition: mean squared error  $\mathbb{E}[|\hat{\theta} - \theta|^2] = |\mathbb{E}[\hat{\theta}] - \theta|^2 + \mathbb{V}(\hat{\theta}) = \text{bias}^2 + \text{variance}$ .
  - How can we interpret the mean squared error: average  $|\hat{\theta} - \theta|^2$  over  $T$  simulations (for large  $T$ ).
  - Minimum Variance Unbiased Estimator: Choose the one with minimum variance.
- (d) Some examples
  - Sample mean  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ , sample variance  $\hat{S}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ : both unbiased estimator.
  - (MLE estimator) Gaussian mean  $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$ , Gaussian variance  $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$ .
  - (MLE estimator) Exponential parameter  $\hat{\lambda} = n / \sum_{i=1}^n X_i$ .

### 2. Maximum Likelihood Estimation

- Intuition: MLE sets the parameter so that the likelihood of seeing the data we observe is the largest.
- Procedure:
  - (a) Write down the likelihood function using the definition of the random sample and  $f(x; \theta)$ .

$$L(\theta; X_1, \dots, X_n) = p(X_1, \dots, X_n; \theta) = \prod_{i=1}^n f(X_i; \theta).$$

$L(\theta; X_1, \dots, X_n)$  is the function of  $\theta$ .

- (b) It might be easier to work with the log-likelihood function (also a function w.r.t.  $\theta$ )

$$\ell(\theta; X_1, \dots, X_n) = \log p(X_1, \dots, X_n; \theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

- (c) Find  $\hat{\theta}$  that maximizes  $L(\theta; X_1, \dots, X_n)$  or  $\ell(\theta; X_1, \dots, X_n)$  (the two are equivalent).
  - i. A standard routine: (1) calculate the first derivative of  $\ell(\theta)$  with respect to  $\theta$ , set it to be zero.
  - (2) show that  $L(\theta)$  attains maximum value at some point.

- **Example 1 (MLE of Shifted Exponential)**

Consider  $n$  independent observations  $x_1, \dots, x_n$  from the population with the shifted exponential pdf:

$$f(x; \lambda, \theta) = \begin{cases} \lambda e^{-\lambda(x-\theta)} & \text{if } x \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

The parameters  $\theta$  and  $\lambda$  are unknown and we would like to obtain estimates. Note that  $\theta, \lambda$  takes values in reals and  $\lambda > 0$ .

- (a) Obtain the maximum likelihood estimators of  $\theta$  and  $\lambda$ .  
 (Hint: First try to find the MLE of  $\theta$  by considering what happens to the likelihood function when  $\min(x_1, \dots, x_n) \geq \theta$  and  $\min(x_1, \dots, x_n) < \theta$  separately. Then find the MLE of  $\lambda$  in terms of the MLE of  $\theta$ ,  $n$ , and  $x_1, \dots, x_n$ .)
- (b) If  $n = 10$  with observed values 3.11, .64, 2.55, 2.20, 5.44, 3.42, 10.39, 8.93, 17.82, and 1.30, calculate the maximum likelihood estimates of  $\theta$  and  $\lambda$  based on your results from the previous part.
- **Solution:** (a) The likelihood function of the distribution for random sample of  $X_1, X_2, \dots, X_n$  is,

$$L(x_1, x_2, \dots, x_n; \lambda, \theta) = \prod_{i=1}^n \lambda e^{-\lambda(x_i - \theta)} = \lambda^n \times e^{-\lambda \sum_{i=1}^n (x_i - \theta)}.$$

Assuming the likelihood with respect to  $\theta$ . According to the pdf of the distribution, for each value of  $i = 1, 2, \dots, n, x_i \geq \theta$ . That is,  $x_1 \geq \theta, x_2 \geq \theta, \dots, x_n \geq \theta$ . If the minimum of  $x_i$  is greater than  $\theta$ , then there is possibility of these condition,

$$\min(x_i) \geq \theta$$

The log likelihood function is computed by taking the logarithm of the likelihood function. That is,

$$\ell(X_1, \dots, X_n; \lambda, \theta) = \ln[f(X_1, X_2, \dots, X_n | \theta)] = \ln\left[\lambda^n \times e^{-\lambda \sum_{i=1}^n (X_i - \theta)}\right] = n \ln(\lambda) - \lambda \sum_{i=1}^n (X_i - \theta).$$

To maximize  $\ell(X_1, \dots, X_n; \lambda, \theta)$ , we want the second term to be as small as possible, equivalently, we need  $\theta$  to be as big as possible. Together with the condition that  $\min(X_i) \geq \theta$ , we have that the MLE of  $\theta$  is

$$\hat{\theta} = X_{(1)} := \min(X_i).$$

Now, we look at  $\lambda$ . Differentiating the log likelihood function with respect to  $\lambda$  and equate it to zero to obtain the maximum likelihood estimator of  $\lambda$ .

Thus,

$$\begin{aligned} \frac{d}{d\lambda} \ln[f(X_1, X_2, \dots, X_n | \hat{\theta})] &= 0 \\ \frac{d}{d\lambda} n \ln(\lambda) - \lambda \sum_{i=1}^n (X_i - \hat{\theta}) &= 0 \\ \frac{n}{\lambda} - \sum_{i=1}^n (X_i - \hat{\theta}) &= 0 \\ \frac{n}{\lambda} &= \sum_{i=1}^n (X_i - \hat{\theta}). \end{aligned}$$

Therefore, the maximum likelihood estimator of  $\lambda$  is  $\hat{\lambda} = \frac{n}{\sum_{i=1}^n (X_i - \hat{\theta})}$ .

(b) From part (a), we have  $\hat{\theta} = \min(X_i) = 0.64$  and

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^n X_i - n \min(X_i)} = \frac{10}{55.8 - (10 * 0.64)} = 0.2024.$$

- **Example 2** Suppose  $X_1, \dots, X_n$  is a simple random sample from Uniform $[0, 2\theta]$  with some unknown parameter  $\theta$ .

- (a) Note that  $E[X_1] = \theta$ . Then it coincides with estimating the mean parameter of the distribution. Consider the mean estimator  $\bar{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$ . Is it an unbiased estimator? Find the bias and the variance for that estimator.

- (b) Find the MLE for  $\theta$ . Is it an unbiased estimator? Calculate the bias and variance for the MLE estimator  $\hat{\theta}$ .
- (c) Which estimator is better (in terms of mean squared error)?

– **Solution:** (a) We have  $\mathbb{E}[\bar{\theta}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \theta$  and

$$\text{Var}[\bar{\theta}] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n} \text{Var}(X_1) = \frac{\theta^2}{3n}.$$

By definition, it is an unbiased estimator.

(b) The density function is  $f(x; \theta) = \frac{1\{0 \leq x \leq 2\theta\}}{2\theta}$  and thus the log-likelihood function given the random sample is

$$\ell(\theta) = \begin{cases} -n \ln(2\theta) & 0 \leq X_1, \dots, X_n \leq 2\theta \\ -\infty & \text{otherwise} \end{cases}$$

It is easy to see that  $L(\theta)$  attains maximum at  $\hat{\theta} = \max_{1 \leq i \leq n} X_i / 2$ .

In order to calculate the bias and variance for  $\hat{\theta}$ , we need to first identify its distribution. We first calculate the CDF for the random variable  $Y = \max_{1 \leq i \leq n} X_i$ . For any  $y \in [0, 2\theta]$ , we have

$$\mathbb{P}[Y \leq y] = \mathbb{P}[\forall i, X_i \leq y] = \prod_{i=1}^n \mathbb{P}[X_i \leq y] = \left(\frac{y}{2\theta}\right)^n.$$

Thus the PDF for the random variable  $Y$  is given by  $f_Y(y) = 1\{0 \leq y \leq 2\theta\} n(2\theta)^{-n} y^{n-1}$ . This implies that

$$\mathbb{E}[Y] = \int_0^{2\theta} n(2\theta)^{-n} y^n dy = \frac{2n}{n+1} \theta \quad \text{and} \quad \mathbb{E}[Y^2] = \int_0^{2\theta} n(2\theta)^{-n} y^{n+1} dy = \frac{4n}{n+2} \theta^2.$$

Given  $\hat{\theta} = \frac{1}{2}Y$ , we have

$$\text{Bias} = \frac{1}{2} \mathbb{E}[Y] - \theta = -\frac{\theta}{n+1} \quad \text{and} \quad \text{Var}[\hat{\theta}] = \frac{\mathbb{E}[Y^2] - (\mathbb{E}[Y])^2}{4} = \frac{n}{(n+2)(n+1)^2} \theta^2.$$

(c) We have

$$\text{MSE}(\bar{\theta}) = \frac{\theta^2}{3n} \quad \text{and} \quad \text{MSE}(\hat{\theta}) = \frac{\theta^2}{(n+1)(n+2)}$$

It is easy to see that  $\text{MSE}(\bar{\theta}) > \text{MSE}(\hat{\theta})$ .  $\hat{\theta}$  converges to  $\theta$  much faster than  $\bar{\theta}$  does. (in order to achieve approximately  $\varepsilon$  accuracy, requires  $\varepsilon^{-1}$  data for  $\bar{\theta}$ , but  $\varepsilon^{-1/2}$  data for  $\hat{\theta}$ .)